

PRINCE2 | P2-15

# Lessons Log

Repository of lessons that apply to this project or future projects

## Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

| Field               | Detail                                    |
|---------------------|-------------------------------------------|
| Document title      | Lessons Log                               |
| Document reference  | P2-15-HEL-T2-v1.0                         |
| Project / Programme | Helios Tranche 2 — Scale Build            |
| Version             | 1.0                                       |
| Status              | Baselined                                 |
| Date                | 18 Sep 2026                               |
| Author              | D. Okonkwo, Programme Manager             |
| Owner               | Project Manager                           |
| Approver            | Project Board reviews at stage boundaries |
| Classification      | Internal                                  |
| Retention           | 7 years from programme closure            |

## Document control

### Version history

| Version | Date        | Author                        | Summary of change                        |
|---------|-------------|-------------------------------|------------------------------------------|
| 0.1     | 04 Jun 2026 | D. Okonkwo, Programme Manager | Initial draft                            |
| 0.2     | 21 Jul 2026 | D. Okonkwo, Programme Manager | Updated following workstream lead review |
| 1.0     | 18 Sep 2026 | D. Okonkwo, Programme Manager | Baselined at the Tranche 2 funding gate  |

### Reviewers

| Name                                   | Role                     | Review scope                              | Date reviewed |
|----------------------------------------|--------------------------|-------------------------------------------|---------------|
| T. Nakamura, Chief Platform Architect  | Chief Platform Architect | Technical content and solution alignment  | 08 Sep 2026   |
| L. Haddad, CISO                        | CISO                     | Security, assurance and residency content | 10 Sep 2026   |
| C. Whitfield, Finance Business Partner | Finance Business Partner | Cost, funding and benefit figures         | 11 Sep 2026   |
| M. Verity, Head of P3O                 | Head of P3O              | Governance standards and completeness     | 15 Sep 2026   |

### Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

| Name                                   | Role                                 | Decision                               | Date        |
|----------------------------------------|--------------------------------------|----------------------------------------|-------------|
| R. Al Mazrouei, Group CTO              | Senior Responsible Owner / Group CTO | Approved                               | 18 Sep 2026 |
| H. Farrell, Director of Infrastructure | Director of Infrastructure           | Approved                               | 18 Sep 2026 |
| L. Haddad, CISO                        | CISO                                 | Approved subject to closure of ISS-003 | 18 Sep 2026 |

**Related products**

| Reference | Product        | Relationship                       |
|-----------|----------------|------------------------------------|
| P2-26     | Lessons Report | Compiled from this log             |
| P2-11     | Plan           | Lessons incorporated into planning |

## How to use this template

| Aspect             | Guidance                                                                                                                                                                       |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Records lessons learned during the project and lessons from previous projects that are being applied, so that improvement is evidenced rather than assumed.                    |
| Framework          | PRINCE2 — Created in Starting up a Project; maintained throughout; drawn on for the Lessons Report                                                                             |
| Created when       | Starting up a Project                                                                                                                                                          |
| Reviewed / updated | At every stage boundary and at closure                                                                                                                                         |
| Owner              | Project Manager                                                                                                                                                                |
| Approver           | Project Board reviews at stage boundaries                                                                                                                                      |
| Format options     | Document, presentation, register entry, or tool record — the content matters, not the medium.                                                                                  |
| Tailoring          | Scale the depth of each section to the scale, risk and complexity of the work. Delete sections that add no value and record the tailoring decision in the governance approach. |

*This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.*

### Quality criteria

*Use these to check the product is fit for purpose before submitting it for approval.*

- Lessons state what happened, why, and what should be done differently — not just what went wrong
- Each lesson is categorised so it can be found by future projects
- Lessons brought in from previous projects are recorded at the start, not only at the end
- Lessons are routed to a body able to act on them

Contents

Document control..... 2

    Version history..... 2

    Reviewers..... 2

    Approval..... 2

    Related products..... 3

How to use this template..... 4

    Quality criteria..... 4

1. Lessons register..... 6

2. Category key..... 8

3. Lessons imported from previous projects..... 9

4. Lessons escalated for organisational action..... 10

## 1. Lessons register

| Ref    | Date        | Category                       | Lesson                                                                                                                                    | What happened / evidence                                                                                                                  | Recommendation                                                                                                               | Applies to                     | Logged by                              | Status                           |
|--------|-------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------|----------------------------------|
| LL-001 | 12 Sep 2026 | Planning and estimating        | Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated VLAN    | Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated VLAN    | Model firmware verification as its own work package with parallel VLAN capacity sized for the shipment rate                  | Schedule development           | R. Al Mazrouei, Group CTO              | Adopted — method change made     |
| LL-002 | 24 Sep 2026 | Governance and decision-making | The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together         | The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together         | Convene a single decision workshop with Finance and business unit heads together, with the sponsor present to break deadlock | Governance and decision-making | H. Farrell, Director of Infrastructure | Adopted — method change made     |
| LL-003 | 02 Oct 2026 | Technical and design           | Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass | Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass | Budget two weeks of vendor performance engineering per hall as a planned activity, not as a remedial response                | Estimating                     | D. Okonkwo, Programme Manager          | Adopted — estimate basis changed |
| LL-004 | 08 Sep 2026 | Security and compliance        | A critical finding on cluster API exposure was discovered late because the test was scheduled after build completion                      | A critical finding on cluster API exposure was discovered late because the test was scheduled after build completion                      | Run a security design review at architecture sign-off and a first penetration test at 30% build, not only at completion      | Assurance planning             | S. Bhatt, Project Manager              | Adopted — assurance plan revise  |

| Ref    | Date        | Category                      | Lesson                                                                                                                                                 | What happened / evidence                                                                                                                               | Recommendation                                                                                                   | Applies to              | Logged by                             | Status                                |
|--------|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------|---------------------------------------|
|        |             |                               |                                                                                                                                                        |                                                                                                                                                        |                                                                                                                  |                         |                                       | d                                     |
| LL-005 | 16 Sep 2026 | Stakeholder and communication | Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced                                | Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced                                | Assess attitude against observed behaviour, and engage those losing autonomy before the decision is communicated | Stakeholder engagement  | L. Haddad, CISO                       | Partially adopted                     |
| LL-006 | 05 Sep 2026 | Procurement and suppliers     | Awarding fabric and platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage | Awarding fabric and platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage | Include supplier concentration explicitly in source selection criteria for packages above AED 10m                | Source selection        | M. Verity, Head of P3O                | Accepted — not yet applied            |
| LL-007 | 22 Jan 2027 | Quality and testing           | Benchmarking against a representative training workload rather than a synthetic test exposed a metadata bottleneck that synthetic tests missed         | Benchmarking against a representative training workload rather than a synthetic test exposed a metadata bottleneck that synthetic tests missed         | Make workload-representative benchmarking a mandatory acceptance criterion for storage                           | Acceptance testing      | T. Nakamura, Chief Platform Architect | Adopted — acceptance criteria changed |
| LL-008 | 11 Sep 2026 | Team and resourcing           | Widening the role to remote-first within GST±3 produced four qualified candidates within three weeks after three months of local search                | Widening the role to remote-first within GST±3 produced four qualified candidates within three weeks after three months of local search                | Test the necessity of location constraints before opening scarce technical roles                                 | Resourcing              | J. Ferreira, Network Lead             | Adopted                               |
| LL-009 | 11 Aug 2026 | Requirements                  | Involving the CISO at architecture option stage rather than at review removed two                                                                      | Involving the CISO at architecture option stage rather than at review removed                                                                          | Include the security architect as a named participant in architecture option                                     | Architecture and design | A. Kowalski,                          | Adopted                               |

| Ref    | Date        | Category                | Lesson                                                                                                                                     | What happened / evidence                                                                                                                   | Recommendation                                                                                                       | Applies to                  | Logged by                                     | Status                         |
|--------|-------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------|--------------------------------|
|        |             |                         | <i>weeks of rework on the network policy model</i>                                                                                         | <i>two weeks of rework on the network policy model</i>                                                                                     | <i>workshops</i>                                                                                                     |                             | <i>Data Centre Lead</i>                       |                                |
| LL-010 | 09 Sep 2026 | Planning and estimating | <i>Fast-tracking the structural order against a design that had not been frozen created a live risk of rework worth around AED 900,000</i> | <i>Fast-tracking the structural order against a design that had not been frozen created a live risk of rework worth around AED 900,000</i> | <i>Where fast-tracking is used, record the design freeze point explicitly and make change to it a Board decision</i> | <i>Schedule compression</i> | <i>P. Raghavan, Platform Engineering Lead</i> | <i>Adopted — control added</i> |

## 2. Category key

| Category                    | Scope                                            |
|-----------------------------|--------------------------------------------------|
| Governance                  | Board effectiveness, decision-making, tolerances |
| Planning & estimating       | Estimation accuracy, dependency management       |
| Requirements & scope        | Definition, change, acceptance                   |
| Quality                     | Reviews, testing, defect patterns                |
| Risk & issue                | Identification, response effectiveness           |
| Resourcing & capability     | Skills, availability, onboarding                 |
| Supplier & commercial       | Contracting, performance, handover               |
| Stakeholder & communication | Engagement, expectation management               |



| Category             | Scope                                |
|----------------------|--------------------------------------|
| Tools & process      | Method, tooling, PMO service         |
| Technical / security | Architecture, controls, environments |

### 3. Lessons imported from previous projects

| Source project   | Lesson                                                                                                                                                 | How applied to this project               | Effective?                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------|
| Helios Tranche 1 | Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated VLAN                 | Yes — reflected in the Tranche 2 schedule | Reportable data incident and loss of regulatory standing        |
| Helios Tranche 1 | The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together                      | Yes — workshop held 24 Sep 2026           | On-call rota cannot be sustained beyond a 5x8 model at handover |
| Helios Tranche 1 | Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass              | Yes — included in the Tranche 2 estimate  | Reportable data incident and loss of regulatory standing        |
| Helios Tranche 1 | A critical finding on cluster API exposure was discovered late because the test was scheduled after build completion                                   | Yes — assurance plan revised              | On-call rota cannot be sustained beyond a 5x8 model at handover |
| Helios Tranche 1 | Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced                                | Partially — engagement plan revised       | Reportable data incident and loss of regulatory standing        |
| Helios Tranche 1 | Awarding fabric and platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage | No — for the next procurement             | On-call rota cannot be sustained beyond a 5x8 model at handover |

## 4. Lessons escalated for organisational action

| Lesson ref                                                          | Escalated to             | Date        | Response / action taken       |
|---------------------------------------------------------------------|--------------------------|-------------|-------------------------------|
| Engage vendor performance engineering for two weeks per hall        | Group Security           | 28 May 2027 | Accepted — no change required |
| Retune CDU set-points and model a secondary heat rejection loop     | Head of P3O              | 25 Jun 2027 | Under review                  |
| Broker all vendor access through the monitored in-country jump host | Finance Business Partner | 18 Sep 2026 | Accepted — method change made |
| Engage vendor performance engineering for two weeks per hall        | P3O Lessons Repository   | 15 Oct 2026 | Accepted — no change required |
| Run two live coolant-leak drills before hall A handover             | Portfolio Board          | 30 Nov 2026 | Under review                  |